

Professional Summary

Senior Data Scientist with over 8.5 years of experience in building AI systems across NLP, Computer Vision, Generative AI and Automation of Agentic workflows. Skilled in applying deep learning and large language models to solve complex problems and deliver scalable solutions. Strong foundation in model development, deployment, and iterative improvement using modern MLOps practices.

Skills

- **Python Libraries:** Scikit-learn, OpenCV, Beautiful Soup, TensorFlow, PyTorch, Hugging Face Transformers
- **Machine Learning & AI:** NLP, Computer Vision, Deep Learning, Generative AI, Fine Tuning Representation and Generation Models, Vector Embeddings, Prompt Engineering
- **LLMs:** Azure GPT-4, Google Gemini, Claude, ViT
- **Frameworks & Tools:** Langchain, CrewAI, Autogen, Spacy, Haystack, HuggingFace, MLFlow, AWS and Azure ML services

Work History

Senior Data Scientist

08/2021 - Current

Indegene – Bengaluru, India (Remote)

AI-Powered Design-to-Code Automation:

- **Product Overview:** Spearheaded data science efforts for an AI product that automates the conversion of PDF-based emailers and e-detail designs into responsive HTML. This tool eliminates manual HTML development, streamlining workflows, and improving productivity across design and development teams.
- **Content Extraction:** Utilized Azure Document Intelligence/AWS Textract to accurately extract text, images, and its metadata from PDFs, ensuring seamless design-to-code translation.
- **Custom Object Detection:** Designed and deployed YOLO v10 models to identify key layout features, such as CTAs, lines, and other design elements.
- **Generative AI Implementation:** Engineered workflows using LLMs for automated HTML-to-MJML conversion. Integrated advanced features like background gradient detection, CTA shape identification, and vertical text property extraction.
- **Layout Structuring:** Built custom algorithms to detect rows and column layouts, providing critical insights to front-end teams for responsive HTML generation.

Impact:

- Achieved a 70% reduction in development time for converting varied PDF complexities into responsive HTML.
- Improved content extraction accuracy for production-ready designs. Adopted by 8-10 enterprise clients, each with 5 or more developers, driving significant productivity gains across teams.

Agentic AI for Multimodal PDF-to-HTML Update Automation:

- Developed an autonomous pipeline using CrewAI and LLMs to process annotated PDFs and update HTML files.
- Implemented agentic reasoning to identify annotation intent in PDF(e.g., text edits, image alignments).
- Integrated multimodal inputs (OCR, bounding box, HTML/CSS) to support structured transformations.

- Designed feedback loops for validation and self-correction.

ICB (Intelligent Content Brain)

- **Project Overview:** Developed an AI-powered tool that processes PDFs (e.g., emailers, e-details) to extract and analyze content intelligently, using advanced vision and NLP techniques. This platform provides intelligent content analysis and actionable insights for end clients.
- **Key Message Tagging:** Fine-tuned a multi-class classification model using BERT to identify and tag critical key messages, enhancing automated content analysis.
- **Image/Text Semantic Modeling:** Implemented transfer learning to build a robust image/text similarity model.
- **Multilingual NER Model:** Utilized SpaCy and Hugging Face frameworks to extract NER in 12+ languages, expanding the tool's global applicability for a diverse clientele.
- **MLOps Implementation:** Designed a comprehensive data science pipeline with data collection, model retraining, and model deployment, leveraging AWS and MLFlow for iterative model updates based on feedback loops.

Impact:

- Achieved 70–80% accuracy across domain-specific models with iterative retraining, ensuring reliable content extraction for enterprise clients.
- Enabled seamless processing of PDFs on demand, supporting multiple clients under subscription models for PDF processing.

Data Scientist

10/2019 - 08/2021

Genpact – Bengaluru, India (Remote)

- **Binary Classification for Bad Debt Estimation:** Developed a binary classification model using ensemble techniques to estimate the probability of customers being a Bad Debt.
- **Impact:** Saved approximately 25% of revenue year over year by proactively managing financial risks.
- **Time Series Modeling for Pharmaceutical Sales Prediction:** Utilized the Holt-Winters model for forecasting units and sales of multiple pharmaceutical brands over 2 years. Improved accuracy by applying better seasonality and trend factors.
- **Impact:** Achieved 10-15% higher accuracy compared to the previous SAS model, improving sales predictions.

Data Engineer

12/2016 - 09/2019

LTI – Bengaluru, India

- Worked on Predictive Modelling for internal stakeholders to estimate SLA for Priority Incidents, Resource Estimation based on SLA which enabled 20-25% better use of Resource Management.
- Led Server Infrastructure Development and Management of Production systems for ETL(informatica)
- Involved in Automation process of Admin Console Monitoring which resulted in savings of manual 15% efforts

Education

PGP-Business Analytics And Business Intelligence: Data Science

01/2019 - 01/2020

Great Learning | University of Texas, Austin - Bangalore

Bachelor of Engineering: Electronics And Communications Engineering

06/2012 - 06/2016

Vemana Institute of Technology - Bangalore

Certifications

AWS Certified Machine Learning Specialty Credential URL: https://www.credly.com/badges/f8f42321-1a9f-4c3e-a72c-22f13cdb803c/public_url

AWS Certified Solution Architect Associate Credential URL: https://www.credly.com/badges/561eaf22-04d0-4a92-937c-08bf8526ca0c/linked_in_profile